

AJKD Atlas of Renal Pathology: Kidney Disease in Primary Sjögren Syndrome

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Clinical and Pathologic Features

Primary Sjögren syndrome is an autoimmune disorder with abnormal B- and T-cell responses to autoantigens, resulting in inflammation and injury to exocrine glands, predominantly lacrimal and salivary glands, with resulting dry eyes and mouth (sicca syndrome). It affects 0.01%-0.1% of the population and occurs primarily from age 50 onward, with a female to male ratio of 9:1. Extraglandular disease happens in around 20% of patients, with joint problems in 30%-50%, and less frequently (5%-10%) cutaneous vasculitis, pulmonary manifestations, and peripheral nervous system manifestations. The frequency of kidney disease varies, from very rare to up to 30% in different series. Kidney disease manifests typically 2-3 years after onset of exocrine gland manifestations as insidious low-grade proteinuria and slowly decreasing kidney function, due to acute or chronic interstitial nephritis, in about two-thirds of those who undergo kidney biopsy. Patients may show distal renal tubular acidosis, and less frequently proximal renal tubular acidosis with full-blown Fanconi syndrome, with glycosuria, aminoaciduria, and low-level tubular proteinuria. With immunosuppression, glomerular filtration rate is maintained long term in those with chronic interstitial nephritis. Of Sjögren patients undergoing kidney biopsy, 5%-30% have cryoglobulinemic glomerulonephritis due to ongoing polyclonal B-cell stimulation, which is associated with worse renal and overall survival.

Light microscopy: Patients may show acute interstitial nephritis with lymphoplasmacytic infiltrate and edema, or chronic interstitial nephritis with similar cellular infiltrate with interstitial fibrosis and tubular atrophy. The infiltrate consists of CD4/CD8 T-cells, B-cells, and plasma cells, with scattered tubulitis.

Glomeruli are unremarkable, unless there is concomitant cryoglobulinemic glomerulonephritis, in which case there are proliferative changes, often with periodic acid-Schiff-positive cryo-plugs.

Immunofluorescence microscopy: In cases with tubulointerstitial nephritis, there are no deposits. Cases with cryoglobulinemic glomerulonephritis show cryoglobulin deposits, most frequently IgM dominant with clonal shift, involving mesangium and capillary walls.

Electron microscopy: In cases with tubulointerstitial nephritis, there are no specific changes. In cases with cryoglobulinemic glomerulonephritis, mesangial and subendothelial deposits with short curvilinear substructure are present.

Etiology/Pathogenesis

The glandular epithelium is an initial target and may also trigger the autoimmune polyclonal B-cell proliferation, with autoantibodies to SSA (anti-Ro) and SSB (anti-La). Triggers may involve virus (with expression of the Toll-like receptors TLR-3 and TLR-7) or physical stress, with subsequent expression on exocrine glandular and kidney epithelium of HLA-DR. Continuous polyclonal B-cell stimulation may lead to induction of cryoglobulins, and in some patients (2%-9%), development of lymphoma, most commonly mucosa-associated lymphoid tissue (MALT) lymphoma.

Differential Diagnosis

Chronic interstitial nephritis may be present in tubulointerstitial nephritis with uveitis syndrome, distinguished by the clinical setting. In sarcoidosis, granulomas are common, and are not present in Sjögren syndrome. In IgG4-related disease, there also are frequent plasma cells, ≥ 10 IgG4 plasma cells per high-power field, but associated with a whorling storiform pattern of fibrosis, the latter not present in Sjögren syndrome. Cryoglobulinemic glomerulonephritis due to other causes, such as hepatitis C virus infection and/or hematologic malignancies, are distinguished by clinical features in combination with laboratory testing to rule out or establish viral infection or a neoplastic disorder of lymphoid cells.

Key Diagnostic Features

- Chronic interstitial nephritis, with infiltrating lymphoplasmacytic infiltrate with occasional tubulitis (in clinical setting of sicca syndrome)

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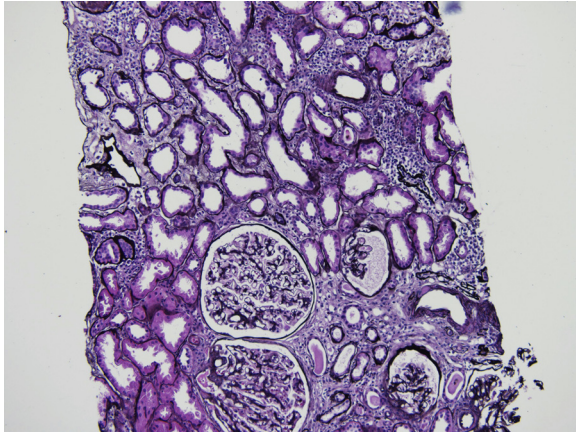


Figure 1. Sjögren syndrome with interstitial lymphoplasmacytic infiltrate involving areas of interstitial fibrosis and tubular atrophy (Jones silver stain).

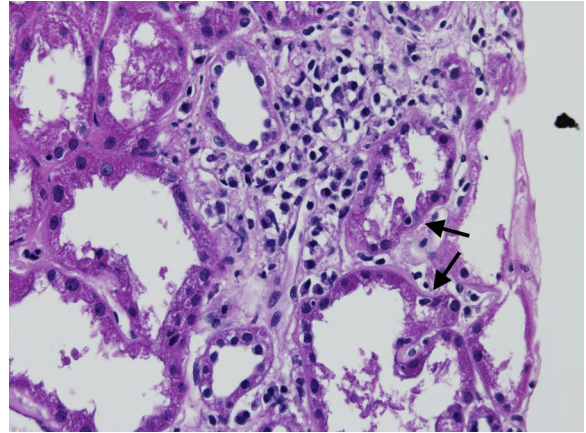


Figure 3. Sjögren syndrome with interstitial lymphoplasmacytic infiltrate with tubulitis (arrows) involving areas of interstitial fibrosis (hematoxylin and eosin stain).

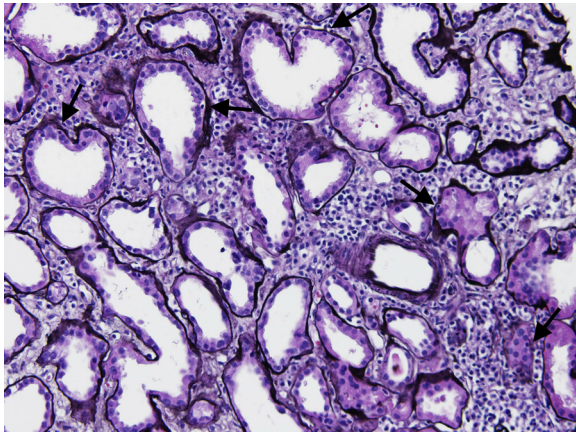


Figure 2. Sjögren syndrome with interstitial lymphoplasmacytic infiltrate with tubulitis (arrows) involving areas of interstitial fibrosis and tubular atrophy (Jones silver stain).

Update

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**Erratum Regarding “AJKD Atlas of Renal Pathology:
Kidney Disease in Primary Sjögren Syndrome”
(*Am J Kidney Dis.* 2017;69[6]:e29-e30)**

In the article entitled “AJKD Atlas of Renal Pathology: Kidney Disease in Primary Sjögren Syndrome” that appeared as an online-only article in the June issue of *AJKD* (Fogo et al, volume 69, issue 6, pages e29-e30), there was an error in the second sentence, where the sex ratio was inadvertently reversed. The sentence has been corrected to read “It affects 0.01%-0.1% of the population and occurs primarily from age 50 onward, with a female to male ratio of 9:1.” This correction was made to the HTML and PDF versions of the article as of June 29, 2017.